5.8 The Proof Is in the Bag!

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. MA.912.LT.4.10 Judge the validity of arguments and given counterexamples to disprove statements.	Learning Targets	• I can prove that two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence. • I can prove that parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can I prove that two triangles are congruent? • How can I prove that parts of triangles are congruent?
Lesson Narrative	In this lesson, students will apply what they have learned about triangle congruency to complete formal proofs. Students will engage in a culminating experience with triangle congruency.

Component Summary		
	5.8.1 Warm-Up (~5 min.)	Two-column proof about triangle congruency *(SE p. 268)*
	5.8.2 Activity (35 min.)	Card sort activity to complete given proofs using given statement and reason cards *(SE p. 268)*
	5.8.3 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 271)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	• Card sort	• Work through each of these proofs prior to facilitating the activity. Students may struggle with certain ones and it will be important that you know the proofs ahead of time in order to provide guidance. • Visit our online teacher area to download a copy of the card sort to prepare for component 5.8.2.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may struggle with ordering the statements. • Student may apply the same marking to multiple sides, causing confusion. • Students may misinterpret or mislabel corresponding parts of triangles. • Students may struggle with matching statements with the correct reasons.	• This activity is a culminating activity that focuses on proofs. With multiple proofs occurring with multiple cards, it would be beneficial to use colored paper for each proof to keep things sorted as you facilitate the activity. • Students may forget this could be different for some statements. It may help to ask questions about whether the statement relies on another statement or if it is independent.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 5.8.2 Activity has students working on multiple different proofs as a group. Consider leveraging the "stronger and clearer each time" protocol to provide a structured and interactive opportunity for students to revise and refine both their ideas and their verbal and written output.	**MA.K12.MTR.5.1: Patterns and Structure** For 5.8.2 Activity, students will be working together to determine the correct steps and reasons to complete proofs that show that two given triangles are congruent, or parts of triangles are congruent. Students will need to create plans to logically order the steps in the proof.
Differentiate Instruction	Consider placing students in groups based on their performance so far in this unit. Students will be applying their knowledge from prior lessons through the process of completing a proof. Students who struggled with triangle congruency will likely struggle proving it. For activities that require you to review proofs (including reviewing the answers), consider using auditory explanations for a proof prior to filling it in. Use common language to describe how the triangles could be proven to be congruent. Then, transition that language into formal geometric language when completing the proof. Consider altering the assignment to eliminate Proof C and Proof F for students who need more time to complete the card sort.	

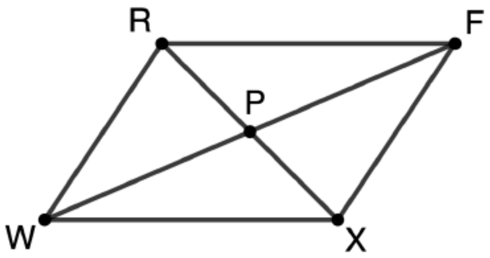
Lesson Notes:

5.8.1 Warm-Up: Bell Work

Component Overview: *Students will complete a two-column proof about triangle congruency. For this activity, students should work individually in order for the instructor to get an idea of where students are entering this content.*

5.8.1 Warm-Up: Bell Work

Polygon $RFXW$ where diagonals $\overline{RX}$ and $\overline{FW}$ bisect each other at point P is shown.



Given: $\overline{RX}$ and $\overline{FW}$ bisect each other at point P.

Prove: $\Delta RPF \cong \Delta XPW$

1. Complete the two-column proof.

Statement	Reason
1. $\overline{RX}$ and $\overline{FW}$ bisect each other at point P	1. Given
2. $RP = PX$ and $WP = PF$	2. Definition of segment bisector
3. $\overline{RP} \cong \overline{XP}$ and $\overline{FP} \cong \overline{WP}$	3. Definition of congruence
4. $\angle RPF \cong \angle XPW$	4. Vertical angles are congruent
5. $\Delta RPF \cong \Delta XPW$	5. SAS Congruence

The data from this Warm-Up can provide data for which students may need additional and immediate support as the next activity launches to ensure they get off on the right track.

5.8.2 Activity: It's in the Bag!

Component Overview: *For this activity, students will work in small groups to complete each of the proofs, using the cards needed for this activity. If students have previously created a list of definitions, postulates, and theorems, allow them to use this during the activity.*

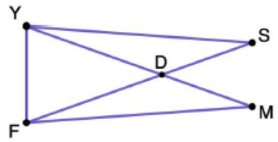
5.8.2 Activity: It's in the Bag!

With your group, complete each proof using the statement and reason cards provided.

Proof A

Given: $\angle YSD \cong \angle FMD$
ΔYDF is isosceles with base $\overline{YF}$

Prove: $\overline{FM} \cong \overline{YS}$

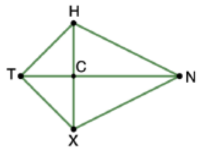


Statement	Reason
$\angle YSD \cong \angle FMD$	Given
ΔYDF is isosceles with base $\overline{YF}$	Given
$\overline{YD} \cong \overline{FD}$	Definition of isosceles triangle
$\angle SDY \cong \angle MDF$	Vertical angles are congruent
$\Delta DMF \cong \Delta DSY$	AAS congruence theorem
$\overline{FM} \cong \overline{YS}$	CPCTC

Proof B

Given: $\overline{TN}$ bisects HTX
$\overline{HT} \cong \overline{XT}$

Prove: $\overline{TN}$ bisects $\angle HNX$



Statement	Reason
$\overline{HT} \cong \overline{XT}$	Given
$\overline{TN}$ bisects $\angle HTX$	Given
$m\angle HTC = m\angle XTC$	Definition of angle bisector
$\angle HTC \cong \angle XTC$	Definition of congruence
$\overline{TN} \cong \overline{TN}$	Reflexive property of congruence
$\Delta TXN \cong \Delta THN$	SAS congruence postulate
$\angle HNT \cong \angle XNT$	CPCTC
$m\angle HNT = m\angle XNT$	Definition of congruence
$\overline{TN}$ bisects $\angle HNX$	Definition of angle bisector

There is a blackline master for this collaborative activity to happen at stations, through a gallery walk, or in traditional small groups.

Prior to the lesson, cut out the statement/reasons from the black-line master and place in plastic bags for students to use when completing proofs. Consider using different-colored paper or other visual indicators for each proof to maintain control over all the correct pieces.

This activity could be done digitally through a platform that provides card sort activities such as Desmos, Padlet, Kahoot etc.

Stronger and Clearer Each Time
Consider leveraging the stronger and clearer each time protocol to provide a structured and interactive opportunity for students to revise and refine both their ideas and their verbal and written output.

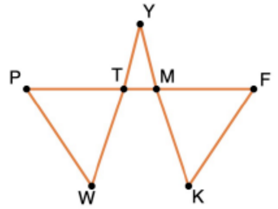
Consider developing a checklist for your use as you observe students working on the proofs.

5.8.2 Activity: It's in the Bag! (continued)

Proof C

Given: $\angle WPT \cong \angle KFM$
$\overline{PW} \cong \overline{FK}$
$\angle YMT \cong \angle YTM$

Prove: $\Delta PWT \cong \Delta FKM$

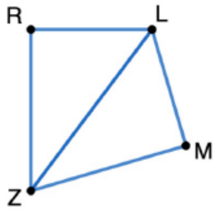


Statement	Reason
$\angle WPT \cong \angle KFM$	*Given*
$\overline{PW} \cong \overline{FK}$	*Given*
$\angle YMT \cong \angle YTM$	*Given*
$\angle YMT \cong \angle FMK$	*Vertical angles are congruent*
$\angle YTM \cong \angle PTW$	*Vertical angles are congruent*
$\angle FMK \cong \angle PTW$	*Transitive property of congruence*
$\Delta PWT \cong \Delta FKM$	*AAS congruence theorem*

Proof D

Given: $\overline{LM} \cong \overline{LR}$
$\overline{LR} \perp \overline{RZ}$
$\overline{LM} \perp \overline{MZ}$

Prove: $\angle MZL \cong \angle RZL$

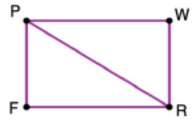


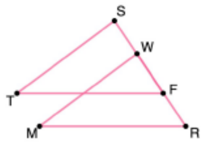
Statement	Reason
$\overline{LR} \perp \overline{RZ}$, $\overline{LM} \perp \overline{MZ}$	*Given*
$\angle LMZ$ and $\angle LRZ$ are right angles	*Definition of perpendicular lines*
ΔLMZ and ΔLRZ are right triangles	*Definition of right triangle*
$\overline{LM} \cong \overline{LR}$	*Given*
$\overline{LZ} \cong \overline{LZ}$	*Reflexive property of congruence*
$\Delta LMZ \cong \Delta LRZ$	*HL theorem*
$\angle MZL \cong \angle RZL$	*CPCTC*

What are some things that you notice that are different about the type of image in Proof C than other images in other proofs you have worked with?

Students often confuse transitive property of congruence and substitution property of equality. Substitution property can apply to statements of equality while transitive property can apply to statements of congruence.

When sharing the correct answers for this activity, consider having groups present their findings, requiring that they also include their explanation and rationale for their answer choices.

Proof E	
Given: $\overline{PF} \cong \overline{WR}$ $\overline{PW} \cong \overline{FR}$ Prove: $\overline{PF} \parallel \overline{WR}$	
Statement	**Reason**
$\overline{PF} \cong \overline{WR}$	*Given*
$\overline{PW} \cong \overline{FR}$	*Given*
$\overline{PR} \cong \overline{PR}$	*Reflexive property of congruence*
$\Delta PRF \cong \Delta RPW$	*SSS congruence postulate*
$\angle FPR \cong \angle WRP$	*CPCTC*
$\overline{PF} \parallel \overline{WR}$	*If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*

Proof F	
Given: $\overline{RF} \cong \overline{SW}$ $\angle TSF \cong \angle MWR$ $\overline{FT} \parallel \overline{MR}$ Prove: $\Delta TSF \cong \Delta MWR$	
Statement	**Reason**
$\overline{RF} \cong \overline{SW}$	*Given*
$RF = SW$	*Definition of congruence*
$FW = FW$	*Reflexive property of equality*
$RF + FW = SW + FW$	*Addition property of equality*
$RF + FW = RW$ and $SW + WF = SF$	*Segment addition postulate*
$RW = SF$	*Substitution property*
$\overline{RW} \cong \overline{SF}$	*Definition of congruence*
$\overline{FT} \parallel \overline{MR}$	*Given*
$\angle TFS \cong \angle MRW$	*If two parallel lines are intersected by a transversal, then corresponding angles are congruent.*
$\angle TSF \cong \angle MWR$	*Given*
$\Delta TSF \cong \Delta MWR$	*ASA congruence theorem*

Consider using some of the time during this activity to pull small groups of students aside to provide additional support based on their performance with this content.

For an additional challenge, consider having students write the proof in a different format, such as a flowchart proof or a paragraph proof.

5.8.3 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

5.8.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about writing proofs.
   Answers will vary.

Consider using the learning targets as sentence frames for students to reflect on using content from the lesson. Alternatively, provide sentence frames specific to a certain concept that you would like additional data to inform differentiation in future lessons.